SFBF-YOLO for Small Object Detection in UAV and Remote Sensing Images

Qiang Hua, Huanzhou Xu, Feng Zhang, Chunru Dong, Boon Han Lim, *Member, IEEE*, and Yong Zhang, *Senior Member, IEEE*

***Abstract*—The small object detection (SOD) in unmanned aerial vehicle (UAV) imagery is fundamentally constrained by the degradation of fine-grained information during hierarchical feature extraction, where conventional downsampling operations induce an inherent representation conflict between high-resolution spatial details and high-level semantic abstraction. To address this, we propose space fine-grained and feature semantics balanced fast YOLO (SFBF-YOLO), an algorithm centered on a novel dual-stream reconstruction mechanism. Instantiated by the space fine-grained attention module (SFGA), this mechanism serves as the architecture's core performance driver. It bifurcates the information flow into a granularity stream that preserves subtle structural attributes, and a contextual stream that models global environmental dependencies, thereby reconstructing a robust feature manifold at the source. Building upon this foundation, we introduce balanced-path aggregation network (BPAN) to enforce a symmetric feature equilibrium principle (SFEP). BPAN explicitly models the bidirectional reciprocity between classification-oriented semantics and localization-oriented precision across scales. Within this synergistic pipeline, a fast ghost-based semantic fusion module (FGSF) executes redundancy-aware feature evolution, ensuring high computational efficiency. Extensive evaluations on VisDrone-2019, USOD, and VEDAI benchmarks demonstrate that SFBF-YOLO achieves a superior tradeoff between precision and efficiency. Notably, on VisDrone-2019, SFBF-YOLO-s achieves a 87% and 47% reduction in parameters and GFLOPs compared with YOLOv8m, while improving mAP_{50} and $mAP_{50\text{-}95}$ by 0.044 and 0.029, respectively. Similarly, SFBF-YOLO-m outperforms YOLOv8l with 79% and 39% lower complexity, alongside accuracy gains of 0.049 and 0.028. These results empirically validate that the proposed dual-stream reconstruction mechanism and SFEP are pivotal for robust detection in complex remote sensing scenarios.**

***Index Terms*—Fine-grained, lightweight network, semantics balanced, small object detection (SOD).**

I. Introduction

SMALL object detection (SOD) is a critical yet challenging task in computer vision, particularly for applications in unmanned aerial vehicle (UAV) and remote sensing imagery [1], [2], [3], [4], [5]. Despite significant progress in general object detection, the performance on SOD tasks often lags. This is primarily because small objects provide scarce informative features and are easily obscured by complex backgrounds or affected by environmental variations (e.g., illumination and fog) and geometric distortions [6], [7]. This results in a significant performance gap compared with detecting normal-sized objects, highlighting the need for specialized detectors.

The research motivation is rooted in two critical, well-established consensuses within the field of UAV and remote sensing observation: the inherent degradation of fine-grained information and the persistent representation conflict between high-resolution spatial details and high-level semantic abstraction in conventional feature pyramids. Regarding the former, conventional hierarchical downsampling via strided convolutions and pooling inevitably leads to the irreversible loss of subtle structural attributes, particularly for tiny targets with limited pixel representation. Simultaneously, a fundamental tradeoff exists where the preservation of these primitive spatial details often occurs at the expense of receptive field expansion, and vice versa. Existing detectors often struggle to reconcile these dual constraints, resulting in either missed detections due to fine-grained feature attenuation or false alarms caused by contextual ambiguity.

To resolve the aforementioned challenges, this study proposes the space fine-grained and feature semantics balanced fast YOLO (SFBF-YOLO) algorithm. The cornerstone of this algorithm is the SFGA module, which adheres to a novel dual-stream reconstruction mechanism. Specifically, during the downsampling process, this mechanism bifurcates the information flow into two specialized pathways: a granularity stream that preserves subtle structural attributes and a contextual stream that models global environmental dependencies. These distinctive streams are subsequently integrated through a synergistic recombination process. Such a design ensures that spatial fine-grained details are revitalized, while contextual information is simultaneously enhanced, providing a robust representation foundation for small targets.

Building upon these enriched feature representations, this study formulates the SFEP to coordinate multiscale interactions. This principle addresses a common limitation, where feature maps in conventional pyramids often exhibit a polarized emphasis on either localization-oriented spatial precision or recognition-oriented semantic abstraction. While most existing fusion networks rely on the unidirectional feature compensation, the SFEP advocates for bidirectional reciprocity, thereby preventing a functional imbalance between localization-driven accuracy and recognition-driven discriminability. This principle is instantiated through the

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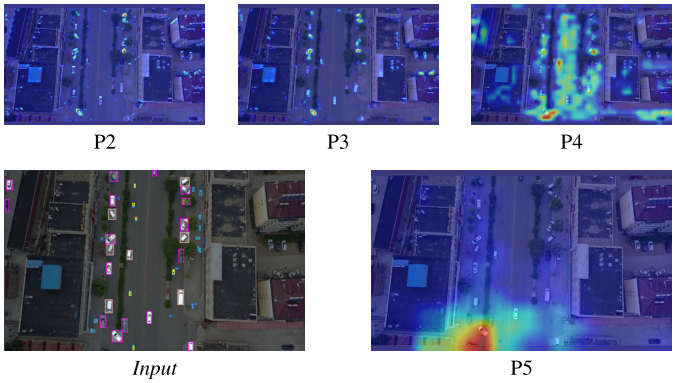


Fig. 1. Heatmaps of multiscale feature maps generated by the backbone network. The deepest layer, P5, shows the minimal activation on the target regions, indicating its limited contribution to SOD.

balanced-PAN (BPAN), which incorporates the fast ghost-based semantic fusion module (FGSF) module as its core redundancy-aware feature evolution engine. By embedding FGSF directly into the dual-path aggregation logic, the BPAN achieves efficient feature evolution and symmetric equilibrium.

Notably, our heatmap visualizations (see Fig. 1) reveal that the deepest feature representations (P5) in conventional pyramids contribute negligibly to small target localization. Guided by this insight, we implement an architectural pruning strategy that eliminates this redundant layer and redistributes computational focus to the high-resolution P2 layer, where the ParNet Block [8] performs final feature refinement.

At last, we present a new model named space fine-grained and feature semantics balanced fast YOLO (SFBF-YOLO). To meet different deployment needs and application scenarios, we provide three model variants: SFBF-n, SFBF-s, and SFBF-m, offering a tradeoff between performance and overhead.

The main contributions of this article are summarized as follows.

1) We propose SFBF-YOLO, a high-performance algorithm tailed for SOD. By integrating a dual-stream reconstruction mechanism and symmetric feature equilibrium principle, it achieves a superior tradeoff between precision and efficiency across three scalable variants (n, s, and m).
2) We formulate the dual-stream reconstruction mechanism, instantiated by the SFGA module. As the core performance driver, SFGA simultaneously preserves intrinsic spatial granularity and expands extrinsic contextual scope, establishing a robust feature foundation at the source.
3) We design the SFEP instantiated by the BPAN. This architecture enforces the bidirectional information reciprocity across feature hierarchies, ensuring that semantic abstraction and spatial precision are reciprocally calibrated.
4) We introduce the FGSF module as a resource-aware efficiency engine. It optimizes the feature evolution process through redundancy mining, significantly reducing computational overhead while maintaining the representational integrity.

The remainder of this article is organized as follows. Section II introduces the related work on small object detection (SOD). Section III details the proposed methodology and the construction of SFBF-YOLO. Experimental results are reported in Section IV. Finally, Section V concludes this article and discusses possible future directions for SOD in UAV and remote sensing imagery.

II. Related Works

This section briefly reviews literature relevant to our work. We first provide an overview of YOLO-based object detection in UAV and remote sensing applications. Then, we survey existing approaches for SOD, including scale-aware and context-modeling. At last, the related lightweight work is reviewed.

A. Applications of YOLO in UAV and Remote Sensing

Deep-learning-based object detectors can be broadly categorized into two-stage detectors and one-stage detectors. Two-stage detectors initiate the detection process by generating potential candidate regions (region proposals) and then employ convolutional neural networks to classify these specified regions. Among the most prominent one-stage detectors is the YOLO series, first proposed by Redmon [9] in 2015. Since its inception, the YOLO series has undergone continuous innovation, with the Ultralytics team recently introducing YOLOv11 [10]. Interestingly, YOLOv8 outperforms other YOLO models in the VisDrone-2019 dataset [11]. Also, YOLOv8 performs even better than many two-stage detectors. We choose YOLOv8 as our baseline. There are many improved YOLO algorithms for SOD of UAV and remote sensing applications.

DMFF-YOLO [12] mitigates the information loss of small objects during downsampling by projecting features into a higher dimensional space through grouped convolutions, thereby preserving original object responses as the network deepens. SOD-YOLO [13] adopts Dysample [14] in the neck to perform upsampling, alleviating the degradation of fine-grained information in shallow features caused by deep feature fusion. PVswin-YOLO [15] integrates Swin Transformers in the backbone to enhance small object saliency against noise, generating higher quality representations. These three methods have been successful in the field of UAVs and remote sensing.

The improved YOLO algorithm primarily addresses the challenges of SOD by focusing on finer-grained image sampling, enhancing the semantic information, and optimizing the feature fusion network. The most significant advantage of such approaches lies in their remarkable overall performance, achieving top-tier detection accuracy while maintaining the lowest parameter count and computational complexity among existing models, thereby facilitating efficient deployment.

B. Scale-Aware Methods

A notable characteristic of SOD tasks is that objects are generally small in size and exhibit significant size variations. The former requires detectors to effectively utilize information from large-scale feature maps, while the latter

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demands detectors capable of handling multiscale feature maps. Early deep learning methods relied solely on high-level features for detection and employed handcrafted techniques (e.g., sliding windows) to address scale variations, resulting in poor performance on SOD tasks. Inspired by the success of reasoning across multilevel in other vision domains [16], [17], researchers have proposed two design principles: designing multibranch architectures or customized training schemes to build detectors tailored to specific scales [18], [19], and fusing hierarchical features to obtain robust, powerful, and discriminative representations of small objects [20], [21], [22].

From the perspective of scale-aware methods, to construct a more efficient small object detector, we removed P5 feature representation, which exhibited weak contribution in Fig. 1, and incorporated the P2 representation into the neck network. Furthermore, a novel BPAN is proposed, where residual connections [23] are intelligently integrated into the baseline PAN architecture to enhance fine-grained details in high-level feature maps while preserving their semantic integrity.

C. Context-Modeling Methods

The contextual information often provides stronger discriminative cues than the target appearance itself, particularly for low-quality imaging scenarios, where the object visibility is severely degraded [24]. In SOD tasks, targets typically suffer from insufficient visual information, necessitating complementary context-object or background interactions for precise localization and classification.

Hu and Ramanan [25] explored encoding regions beyond object boundaries and modeling local context in a scale-invariant manner for tiny face detection. PyramidBox [26] enhanced the discovery of small and blurry faces by leveraging contextual cues. FS-SSD [27] utilized implicit spatial context, such as distances between intraclass and interclass instances, to redetect low-confidence objects. To address the structural distortion caused by RoI pooling, SINet [28] proposed a context-aware RoI pooling layer to preserve the critical contextual information.

CM-YOLO [29] enhances the deep feature representations by aggregating contextual cues and proposes a novel MCB detector head with large-kernel convolutions to capture spatial context, demonstrating success in ship detection. FFCA-YOLO [30] proposed a novel SCAM to model global-region contextual relationships for small objects, achieving state-of-the-art (SOTA) performance in remote sensing applications.

Motivated by these principles, we propose SFGA to explicitly preserve boundary geometry at context-object intersections, while employing large-kernel sparse convolutions to extract spatial contextual information. This enhances boundary texture fidelity and contextual awareness.

D. Lightweight Work

Lightweight design plays a pivotal yet understated role in object detection tasks within UAV and remote sensing domains, driven by the constraints of onboard deployment and real-time inference requirements. In many cases, lightweight models are also synonymous with efficiency, often achieving superior accuracy-to-parameter or accuracy-to-computational ratios, which can be viewed as an alternative form of performance enhancement.

Lightweight approaches is achieved through three integrated strategies: architectural refinement, including the use of highly efficient backbones (e.g., reparameterized blocks [31]), customized feature fusion necks and knowledge distillation, specifically hierarchical matching KD (e.g., reparameterized blocks [32]) applied to high-resolution FPN layers, which ensures the fine-grained small object knowledge is effectively transferred to the compact student models, often resulting in performance superior to the larger teacher networks.

ShuffleNet [33] and GhostNet [34] serve as foundational architectures among traditional highly efficient backbones, which incorporate sophisticated techniques dedicated to increasing information richness. ShuffleNet combines grouped convolutions with channel shuffling, whereas GhostNet approximates standard convolutions through a hybrid operation comprising a reduced standard convolution and a compensatory grouped convolution. Chen et al. [35] argue that the lightweight nature of grouped convolutions increases both training and inference time, and thus propose partial convolutions as an alternative solution.

These advancements motivate the integration of partial convolutions with the ghost paradigm, leading to the fast-ghost convolution. This operator is then utilized to construct the FGSF. This provides a lightweight alternative for YOLO-based models.

E. Summary and Motivation

In summary, existing UAV detection methods are fundamentally bottlenecked by insufficient context modeling, inadequate scale awareness, and computational redundancy. Rather than heuristically stacking modules, we propose SFBF-YOLO as a unified algorithm to explicitly resolve these issues. Specifically, the SFGA module tightly couples fine-grained sampling with global context modeling; BPAN enforces a scale-aware symmetric feature equilibrium; and FGSF executes redundancy-aware feature evolution for lightweight deployment. This synergistic architecture is detailed in Section III.

III. Proposed Method

A. Overview

We select YOLOv8 as our benchmark framework due to its superior performance over other YOLO variants on the VisDrone dataset. This section details the SFGA, FGSF, and BPAN methods sequentially. The overall SFBF-YOLO architecture is then presented. Notably, our modified algorithm prunes YOLOv8's deepest feature representations while reassigning detection tasks to shallow P2 features, and further enhances the pipeline by inserting a Parnet Block prior to the P2 detection head for hierarchical feature enhancement.

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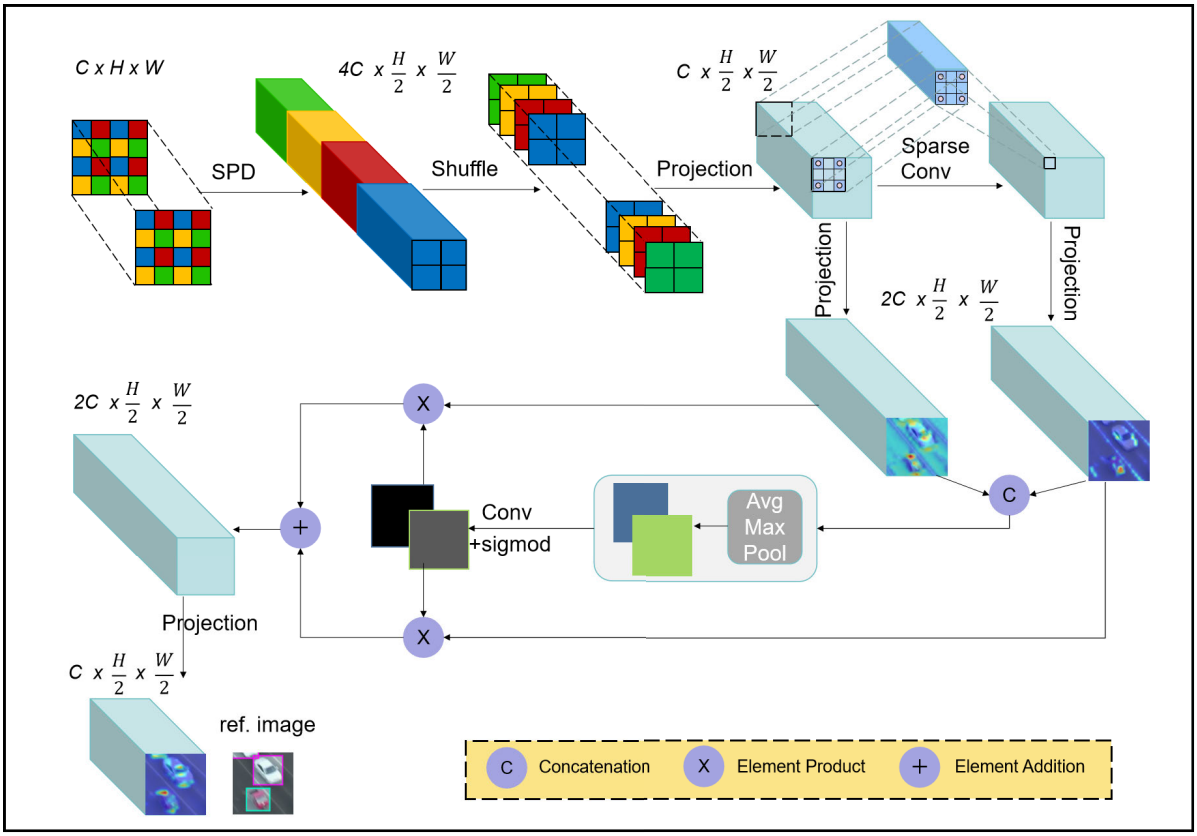


Fig. 2. Structure of the SFGA module. The input feature map is bifurcated into a granularity stream and a contextual stream. These streams are dynamically fused via a selective kernel attention mechanism to reconstruct a high-quality representation for the algorithm.

B. Space Fine-Grained Attention Module (SFGA)

The fundamental challenge in SOD arises from the degradation of fine-grained information inherent in hierarchical feature extraction. As analyzed in Section II-C, traditional context-modeling approaches often operate on downsampled feature maps, where the target's structural attributes have already been compromised. Crucially, when these primitive structural attributes are severely degraded, the efficacy of contextual modeling is inherently diminished, as environmental dependencies lack a well-defined anchor for relational reasoning. This observation implies that an ideal feature extractor for small objects must resolve this conflict by simultaneously ensuring: 1) high granularity preservation and 2) contextual scope expansion.

To this end, we propose the SFGA module, formulated as a dual-stream reconstruction mechanism. As illustrated in Fig. 2, the information flow is bifurcated into two parallel pathways.

Granularity Stream ($\mathbf{x_1}$)*:* This stream aims to lock the target's primitive structural attributes. It utilizes a space-to-depth (SPD) transformation [36] that reindexes spatial pixels into the channel dimension. Unlike lossy pooling or strided convolutions, this pathway ensures preservation of pixel-level granularity, providing a stable spatial foundation even for objects occupying only a handful of pixels.

Contextual Stream ($\mathbf{x_2}$)*:* This stream addresses the semantic scarcity of small objects by employing sparse convolutions with expanded receptive fields. It captures the extrinsic interactions between the target and its local environment, which are crucial for distinguishing actual targets from background clutter.

Synergistic Recombination ($\mathbf{y}$)*:* The recombination logic utilizes an SK-attention subnetwork [37] as a dynamic coordinator. It reconstructs a unified representation by adaptively weighting the fine-grained anchors and contextual priors, ensuring that the final feature manifold is both spatially precise and semantically discriminative.

The SFGA (see Fig. 2) could be written as follows:

$$\mathbf{x_1} = \mathbf{W}\left(\mathrm{CS}\left(\mathrm{SPD}\left(\mathbf{x}\right)\right)\right) \tag{1}$$

$$\mathbf{x_2} = \text{Sparse Conv}\left(\mathbf{x_1}\right) \tag{2}$$

$$\mathbf{y} = \mathrm{SK}\left(\mathbf{x_1}, \mathbf{x_2}\right) \tag{3}$$

where $\mathbf{x} \in \mathbb{R}^{C\times H\times W}$ denotes the input. SPD(·) means space-to-depth operation of SPD convolution [36]. By setting the downsampling scale as S, the feature map is divided into HW/S^2 pixel patches, and every patch get S^2 pixels. By downsampling S^2 times, each time sampling different positions within the patches, the $\mathrm{SPD}(\mathbf{x}) \in \mathbb{R}^{H/S\times W/S\times S^2C}$ is eventually obtained. Actually, in SFBF-YOLO, we set scale = 2.

Also, CS is an abbreviation for channel shuffle, $\mathbf{W}$ denotes the trainable projection matrix, and $\mathbf{W} \in \mathbb{R}^{C\times S^2C}$. The Sparse Conv is sparse convolution, namely, with 5×5 kernel, set expansion ratio = 2 and the number of output channels equal to the number of input channels. SK(·) denotes the selective

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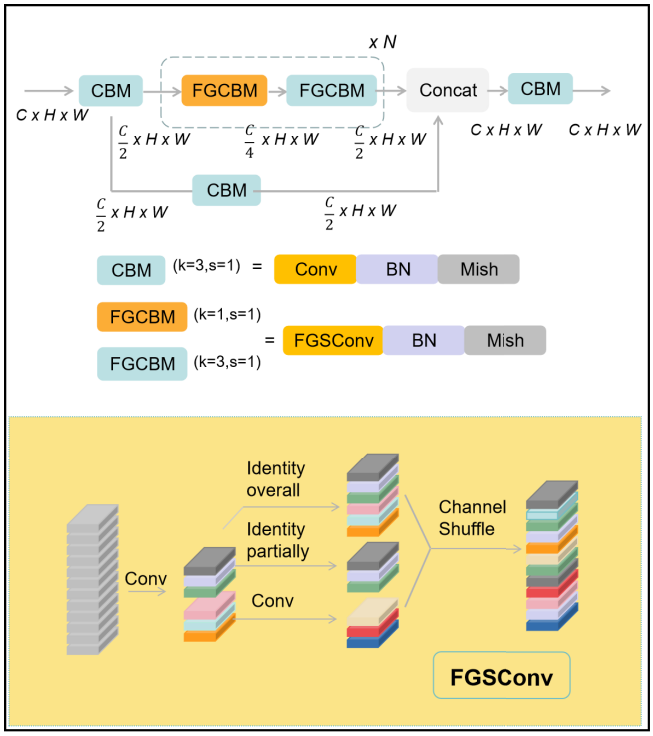


Fig. 3. Structure of FGSF. It provides an efficient and expressive feature mining mechanism that complements the proposed bidirectional pyramid alignment strategy.

kernel subnetwork. For input $\mathbf{x_1}$ and $\mathbf{x_2}$, SK(·) could be expressed as follows:

$$\mathbf{x_3} = \text{Concat}\left(\mathbf{W_1 x_1}, \mathbf{W_2 x_2}\right) \tag{4}$$

$$\mathbf{t} = \text{Sigmoid}\left(\text{Conv}_{5\times5}\left(\text{Concat}\left(\text{AP}\left(\mathbf{x_3}\right), \text{MP}\left(\mathbf{x_3}\right)\right)\right)\right) \tag{5}$$

$$\text{SK}\left(\mathbf{x_1}, \mathbf{x_2}\right) = \mathbf{W_3}\left(\mathbf{T_1} \odot \mathbf{W_1 x_1} + \mathbf{T_2} \odot \mathbf{W_2 x_2}\right) \tag{6}$$

where $\mathbf{W_1}$, $\mathbf{W_2} \in \mathbb{R}^{2C\times C}$ and $\mathbf{W_3} \in \mathbb{R}^{C\times 2C}$ are trainable projection matrices. AP and MP mean the average pooling and max pooling, respectively. $\text{Conv}_{5\times5}$ is standard convolution with 5×5 kernel and set padding $= 2$. $\odot$ denotes the Hadamard product. It should be noted that $\mathbf{t} = [\mathbf{t_1}, \mathbf{t_2}]$, and $\mathbf{t_1}$, $\mathbf{t_2} \in \mathbb{R}^{H/S\times W/S}$. $\mathbf{T_1}$ is formed by concatenating $2C$ instances of $\mathbf{t_1}$, and $\mathbf{T_2}$ is formed by concatenating $2C$ instances of $\mathbf{t_2}$, which is not expressed in mathematical form for the sake of brevity.

Consistent with the principle that high-dimensional feature spaces provide more degrees of freedom for special information recovery, we project the dual-stream feature maps into an expanded latent space. This compensates for the inherent scarcity of discriminative cues in small targets, allowing for more robust feature excavation before the final integration. Subsequently, to synthesize the complementary information from both streams, the projected features are aggregated and subjected to dual-statistic distillation (via parallel max and average pooling). This process condenses the high-dimensional data into compact channel-global descriptors, which capture both the salient feature responses and the background distribution statistics to guide the generation of the final attention weighting matrices. These feature maps are processed through standard convolution and a sigmoid activation to generate spatial weight matrices $\mathbf{T_1}$ and $\mathbf{T_2}$. The matrices perform broadcasted Hadamard products with $\mathbf{W_1 x_1}$ and $\mathbf{W_2 x_2}$, respectively, followed by element-wise summation. This operation essentially reweights complementary feature maps in high-dimensional space through positionwise recalibration, where $\mathbf{T_1}$ and $\mathbf{T_2}$ dynamically adjust the spatial importance of discriminative regions while suppressing noise. Following the reweighting operation, the feature dimensions undergo inverse projection to restore their original channel cardinality. Eventually, we get the output $\mathbf{y} \in \mathbb{R}^{C\times H/2\times W/2}$.

Overall, the SFGA module functions as a unified information reconstructor: the granularity stream enriches fine-grained structural attributes, while the contextual stream captures extrinsic semantic dependencies. Their dynamic fusion ensures complementary synergy throughout the feature extraction process.

C. Fast Ghost-Based Semantic Fusion Module

To enable the effective deep feature mining under strict computational constraints, we introduce the FGSF module, which is designed as a resource-aware feature excavation operator rather than a lightweight structural replacement. The core motivation of FGSF is to balance the computational sparsification and information integrity preservation when handling high-dimensional representations required for SOD.

FGSF is inspired by the observation that both sufficient network depth and high-dimensional feature spaces are essential for extracting weak semantic cues from small objects; yet, conventional dense convolutions incur substantial redundancy that limits real-time deployment. To address this issue, FGSF follows a ghost redundancy evolution principle by integrating partial convolution with the ghost paradigm, enabling adaptive feature refinement under limited resources.

As illustrated in Fig. 3, FGSF operates through a compact three-stage process. First, a convolution compresses spatial features to suppress background noise while retaining target-relevant responses. Next, partial convolution is employed to extract representative semantics, while the remaining channel dimensions are efficiently replenished as ghost features. Finally, the synthesized features undergo channel shuffling. This operation breaks the segregation between intrinsic and generated channels, facilitating the uniform information flow across the entire feature manifold to prevent feature isolation.

By explicitly modeling redundancy-aware semantic evolution and coarse-to-fine feature coupling, FGSF provides an efficient and expressive feature mining mechanism that complements the proposed bidirectional pyramid alignment strategy.

FGSF (see Fig. 3) could be expressed as follows:

$$\mathbf{u_0} = \text{Conv}\left(\mathbf{v}\right) \tag{7}$$

$$\mathbf{u_i} = \text{FGSC}\left(\text{FGSC}\left(\mathbf{u_{i-1}}\right)\right), \quad (i = 1, 2, 3, \ldots, N) \tag{8}$$

$$\mathbf{z} = \text{Conv}\left(\text{Concat}\left(\text{Conv}\left(\mathbf{u_0}\right), \mathbf{u_N}\right)\right). \tag{9}$$

The floating point operations (FLOPs) of FGSFEM can be calculated as follows:

$$r_f \approx (0.8N + 2)\left(HWC^2\right). \tag{10}$$

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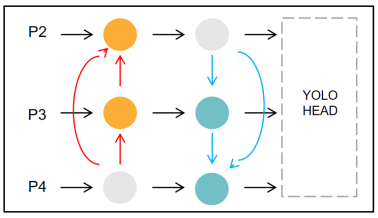


Fig. 4. Structure of SFEP. Unlike unidirectional feature fusion, SFEP explicitly models a bidirectional reciprocity. The enhancement in the top-down path is responsible for propagating high-level semantic abstraction to ensure the sufficient category discrimination, while the enhancement in the bottom-up path focuses on preserving and reinforcing spatial structure to maintain precise localization cues.

Also, the FLOPs of C2F module can be calculated as follows:

$$r_c = (5n + 2)\left(HWC^2\right) \tag{11}$$

where N and n mean the quantity of bottleneck in FGSFEM and C2F. Actually, n equals 3 in YOLOv8 and we set $N = 1$ in SFBF-YOLO

$$\frac{r_f}{r_c} \approx \frac{1}{6}. \tag{12}$$

The proposed FGSF module offers a lightweight alternative to the feature enhancement block within the improved YOLO algorithm. The FGSF is specifically designed to maximize the efficiency of the enhancement module by integrating the respective advantages of ghost convolution, partial convolution, and the channel shuffle operation.

D. Balanced-PAN

In feature pyramid networks, multiscale interaction is traditionally formulated as a feature fusion problem, where high-level semantic information is propagated downward to enhance shallow features, or low-level spatial details are injected upward to compensate deeper representations. Such designs implicitly assume that information enhancement in a single-dominant direction is sufficient for improving the detection performance.

However, in practical UAV scenarios, objects exhibit an extensive scale distribution where extremely tiny targets and moderately small objects often coexist. Purely unidirectional compensation tends to induce a functional imbalance: excessive semantic injection may suppress the subtle structural attributes critical, while overemphasizing spatial details may weaken the discriminability.

From this perspective, we reformulate multiscale interaction as a feature balance modeling strategy rather than conventional feature fusion. Based on this logic, we propose the BPAN, which follows SFEP (as shown in Fig. 4) to explicitly model balanced semantic complementarity across feature hierarchies.

Unlike conventional bidirectional pyramid designs, BPAN assigns asymmetric and complementary roles to the two propagation directions. Specifically, the top-down path is responsible for propagating high-level semantic abstraction to ensure the sufficient category discrimination, while the bottom-up path focuses on preserving and reinforcing spatial structure to maintain precise localization cues. By explicitly coordinating these two roles, BPAN resolves the conflict between semantic abstraction and spatial details in a balanced manner. To ensure this equilibrium is achieved efficiently, BPAN incorporates the FGSF module as its core feature evolution engine, which executes redundancy-aware fusion within the bidirectional aggregation logic.

Algorithm 1 Synergistic Forward Pass of SFBF-YOLO

Require: Input UAV/Remote Sensing image $I \in \mathbb{R}^{3\times H\times W}$.
Ensure: Multi-scale detection maps $\mathcal{D} = \{(P^l_{cls}, P^l_{reg})\}^4_{l=2}$. {The detection outputs $\mathcal{D}$ consist of classification maps P_{cls} and regression maps P_{reg} across three scales}

Initialize: Backbone B, Neck N, Detection Head H.
// Step 1: Feature Extraction with SFGA
$f_0 \leftarrow \text{CBS}_{3\times3}(I)$; {Initial stem}
$P2 \leftarrow \text{CBS}(\text{C2F}(\text{SFGA}_1(f_0)))$;
$P3 \leftarrow \text{CBS}(\text{C2F}(\text{SFGA}_2(P2)))$;
$P4 \leftarrow \text{CBS}(\text{C2F}(\text{SPD-S}(P3)))$;
// Step 2: BPAN
// Top-down
$P3_{td} \leftarrow \text{FGSF}((\text{Upsample}(P4))\text{Concat}(P3))$;
$P2_{td} \leftarrow \text{FGSF}((\text{Upsample}(P3_{td}))\text{Concat}(P2) \oplus P4)$;
// Bottom-up
$P3_{mid} \leftarrow \text{FGSF}((\text{FGSConv}(P2_{td}))\text{Concat}(P3_{td}))$;
$P3_{out} \leftarrow P3_{mid} \oplus P2_{td}$;
$P4_{out} \leftarrow \text{FGSF}((\text{FGSConv}(P3_{out}))\text{Concat}(P4))$;
// Step 3: Semantic Refinement and Detection
$P2_{out} \leftarrow \text{ParNetBlock}(P2_{td})$; {Enhancement for P2}
$\mathcal{D} \leftarrow \text{Head}(P2_{out}, P3_{out}, P4_{out})$;
return $\mathcal{D}$

As illustrated in Fig. 5, BPAN performs bidirectional feature interaction through a dedicated complementation operation. In the top-down pathway, higher resolution feature maps receive semantic guidance from deeper layers, enabling shallow features to enhance recognition capability without overwhelming spatial details. Conversely, in the bottom-up pathway, deeper feature maps are complemented with fine-grained spatial information from shallower layers, mitigating semantic over-smoothing and improving localization awareness. This bidirectional reciprocity establishes a stable and balanced information flow across scales.

In this work, residual connections are adopted as a concrete instantiation of the complementation operation due to their efficiency and stability. It is worth noting that the proposed SFEP is implementation agnostic and is not restricted to residual formulations. Alternative feature reconciliation strategies, such as DySample, concatenation-based fusion, deconvolution, or other lightweight modulation mechanisms, can also be incorporated within the SFEP. The core contribution of BPAN, therefore, lies not in the specific operation itself, but in the principled modeling of bidirectional feature equilibrium.

By shifting the focus from feature fusion to equilibrium modeling, BPAN provides a generic and extensible design principle for multiscale representation alignment, which can be seamlessly integrated into different detection architectures and backbones.

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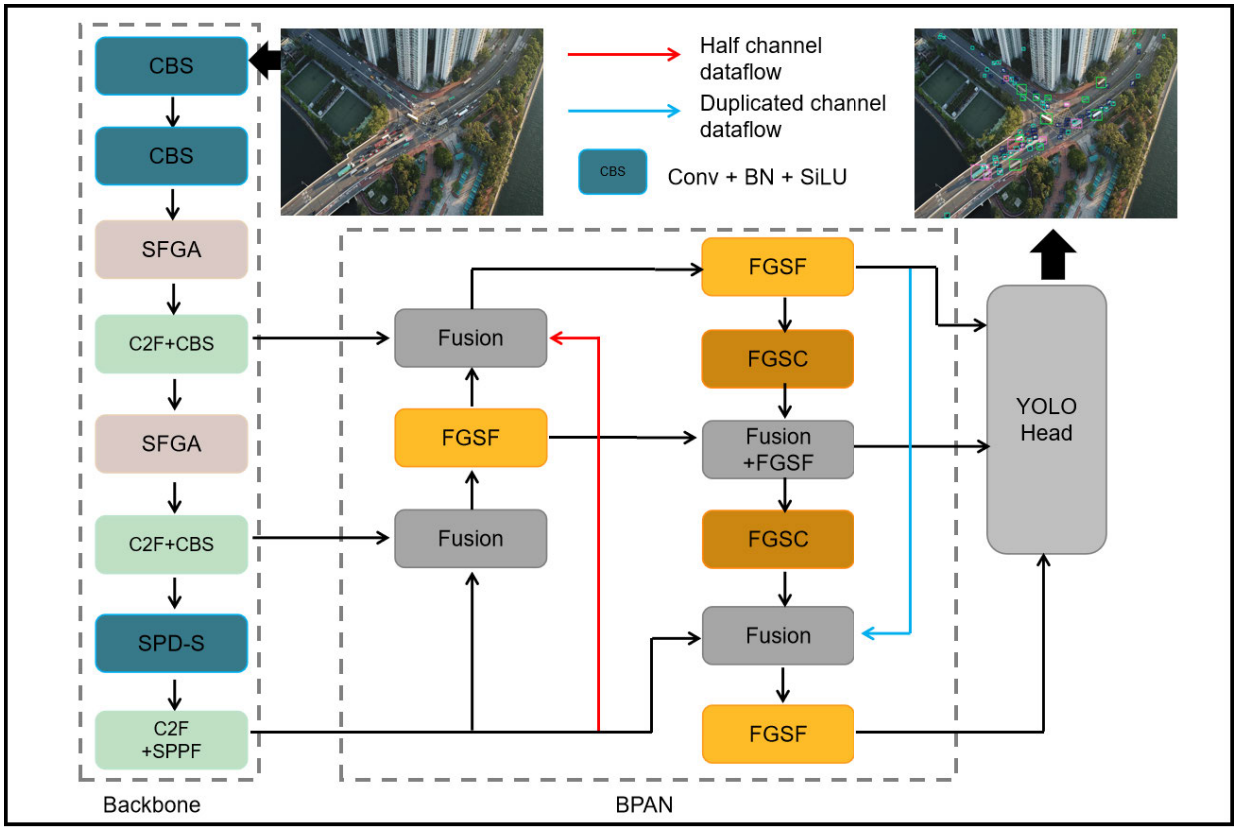


Fig. 5. Overall architecture and detailed forward inference logic of the proposed SFBF-YOLO.

E. SFBF-YOLO

The proposed SFBF-YOLO is built upon a tightly coupled interaction among SFGA, BPAN, and FGSF, forming a unified algorithm rather than a collection of independent enhancement modules. Each component addresses a specific aspect of SOD, while their cooperation enables robust and efficient feature representation.

At the global level, SFGA serves as a fine-grained preservation and semantic guidance mechanism that selectively emphasizes object-relevant regions and suppresses background interference. By providing spatially aware semantic priors, SFGA establishes favorable conditions for subsequent multi-scale feature alignment. These guided features are then fed into BPAN, which explicitly models bidirectional semantic complementation across pyramid levels. Instead of unidirectional feature enhancement, BPAN performs role-aware semantic coordination, ensuring balanced propagation of semantic abstraction and spatial precision across scales.

Within each pyramid level, FGSF further refines the aligned features through redundancy-aware semantic evolution under limited computation. By selectively excavating informative channels while preserving information integrity, FGSF enhances local semantic expressiveness without disrupting the balanced feature distributions established by BPAN. In this way, BPAN defines how semantic information should be coordinated across scales, while FGSF determines how efficiently such information is refined within each scale.

Overall, SFGA provides a powerful representation foundation, BPAN enforces cross-scale feature balance, and FGSF ensures efficient deep feature mining. Their coordinated interaction enables SFBF-YOLO to maintain robust detection performance while satisfying real-time constraints, forming an integrated and extensible algorithm.

IV. Experiments

In this section, we present comprehensive experimental results to validate the effectiveness of the proposed method. Benchmark evaluations are conducted on four representative datasets: VisDrone-2019 [11], USOD [30], and VEDAI [38]. We do comprehensive experiments on VisDrone-2019 and USOD. Given that YOLOv8 consistently outperforms other YOLO variants on the VisDrone dataset, it is adopted as the baseline detector throughout our experiments. Fig. 6 concisely shows the performance comparison of SFBF-YOLO and other methods on the VisDrone-2019 dataset.

The evaluation metrics include mean average precision at an intersection over union (IoU) threshold of 0.5 (mAP_{50}), the expected mean average precision over IoU thresholds from 0.5 to 0.95 ($mAP_{50\text{-}95}$), as well as the number of parameters and FLOPs of the models. The experimental environment is built on Python 3.8 (Ubuntu 20.04), PyTorch 1.11.0, and CUDA 11.3. The hardware configuration includes a 14 vCPU Intel[1] Xeon[1] Platinum 8362 CPU@2.80 GHz. An NVIDIA RTX 3090 GPU (24 GB) is used for experiments on the VisDrone dataset, while two RTX 3090 GPUs (24 GB) are utilized for experiments on the other datasets.

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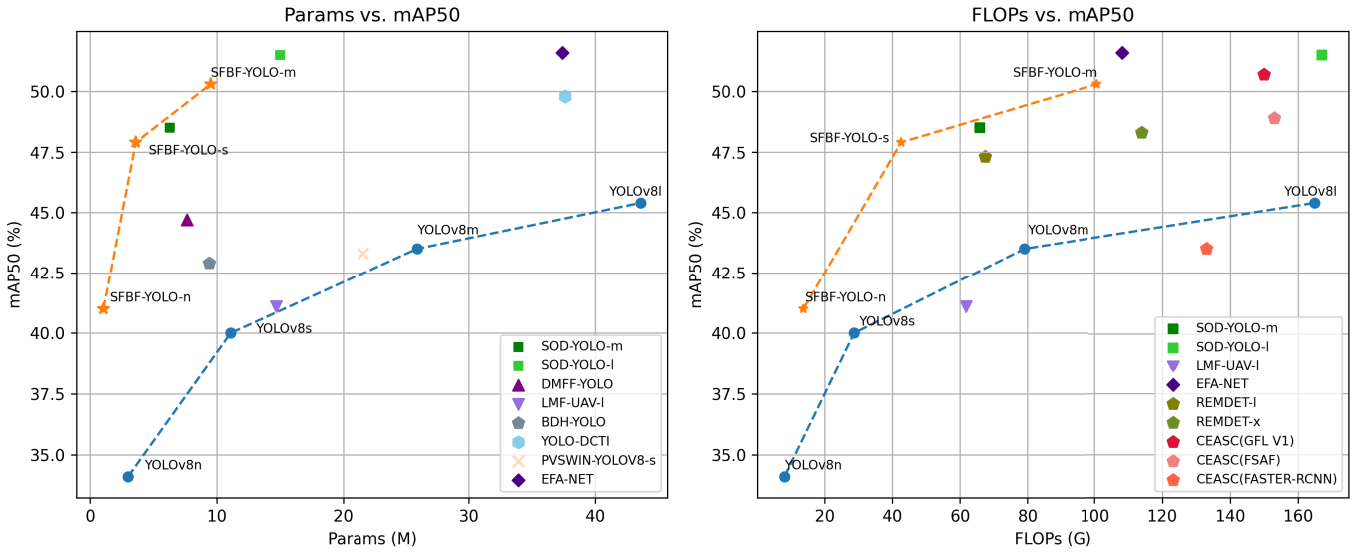


Fig. 6. Performance comparison of SFBF-YOLO and other methods on the VisDrone-2019 dataset. SFBF-YOLO is our model and YOLOv8 is the baseline. The scattered points indicate other improved YOLO-based detectors, as well as representative real-time and two-stage SOTA object detection models.

TABLE I
COMPARISON EXPERIMENTS FOR SFBF-YOLO IN VISDRONE-2019

Model	Para(M)	Gflops	mAP_{50}	mAP_{50-95}	FPS
SFBF-YOLO-n	1.06	13.7	0.41	0.243	66
SFBF-YOLO-s	3.61	42.6	0.479	0.292	53
SFBF-YOLO-m	9.52	100.6	0.503	0.307	48
YOLOv8n	3.0	8.1	0.341	0.195	90
YOLOv8s	11.1	28.7	0.4	0.238	54
YOLOv8m	25.9	79.1	0.435	0.263	49
YOLOv8l	43.6	164.9	0.454	0.279	39
YOLOv10-s [39]	8.1	24.8	0.389	0.233	-
YOLOv9-c [40]	50.9	237.8	0.481	0.299	38
YOLOv11-s [10]	9.4	-	0.395	0.236	-
SOD-YOLO-m [13]	6.3	65.8	0.485	0.295	40
SOD-YOLO-l	17.6	167	0.515	0.32	38
UAV-YOLO [41]	10.3	-	0.47	0.292	46
DMFF-YOLO [12]	7.65	-	0.447	0.272	33
LMF-UAV(s) [42]	6.3	17.6	0.364	0.213	56
LMF-UAV(l)	14.7	61.8	0.411	0.246	-
EFA-Net [43]	37.4	108.2	0.516	-	48
BDH-YOLO [44]	9.39	-	0.429	0.262	-
PVswin-YOLOv8-s [15]	21.6	-	0.433	0.264	68
YOLO-DCTI [45]	37.6	-	0.498	0.274	-
PC-YOLO11s [46]	7.1	-	0.438	0.263	47
RemDet-L [47]	-	67.4	0.473	-	-
RemDet-X	-	114	0.483	-	-
CEASC(GFL V1) [48]	-	150	0.507	-	21
CEASC(FSAF)	-	153	0.489	-	19
CEASC(Faster-RCNN)	-	133	0.434	-	21

In all experiments, we set the learning rate to 0.01 and use the SGD optimizer with a weight decay of 0.0005. Specifically, the input image size is uniformly set to 640×640 across all experiments, a resolution that inevitably exacerbates the degradation of fine-grained information. This stringent condition is applied to rigorously validate the proposed method's robustness and effectiveness, aligning with standard practices in the field. Each model is trained with a batch size of 8 for 300 epochs. No pretrained parameters are used in any of the experiments.

A. Experimental Dataset

VisDrone-2019: As a premier benchmark for SOD in drone-based scenarios, the VisDrone-2019 dataset was collected by

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the AISKYEYE team at Tianjin University. The dataset comprises instances captured across 14 different cities, ensuring a high degree of environmental diversity. It covers a wide spectrum of urban and suburban scenes, including residential communities, highways, and crowded public squares, under varying weather conditions (e.g., sunny, cloudy, and rainy) and lighting intensities.

Structurally, the dataset is partitioned into 6471 images for training, 548 for validation, and 1610 for testing. It features ten specific object categories, including pedestrians, people, cars, vans, buses, trucks, motorbikes, bicycles, awning-tricycles, and tricycles. Notably, VisDrone-2019 is characterized by its high-resolution imagery (up to 2000×1500 pixels) and extreme variations in object scale. The distinct lack of large targets, combined with heavy occlusion, motion blur, and dense object clustering, presents significant challenges for detecting small-scale targets in complex aerial environments.

USOD: To evaluate the model's performance on extremely small-scale targets, we utilize the USOD dataset introduced by Zhang et al. [30]. Derived from the large-scale UNICORN2008 database, USOD is a dedicated benchmark for vehicle detection in high-resolution optical imagery with a ground sample distance of approximately 0.4 m. The dataset construction involved a rigorous process of filtering visible light data, segmenting specific Regions of Interest, and manually annotating vehicle instances to ensure high-label quality.

The dataset consists of 3000 images containing a total of 43 378 annotated targets. It follows a standard random split ratio of 7:3, resulting in 2100 images for training and 900 for testing. The most distinct characteristic of USOD is the minuscule size of its targets: notably, 96.3% of the object instances is smaller than 16×16 pixels and 99.9% fall below 32×32 pixels. This makes USOD one of the most challenging benchmarks for testing the limits of SOD algorithms. Furthermore, the dataset incorporates complex scenarios featuring dense object clustering, severe occlusion, and low-illumination environments, effectively simulating real-world remote sensing difficulties.

Vehicle Detection in Aerial Imagery (VEDAI): The VEDAI dataset [38] comprises cropped samples derived from the large-scale Utah Automated Geographic Reference Center database. The original aerial orthophotos were acquired at a consistent altitude, offering a high spatial resolution of approximately 12.5×12.5 cm per pixel. Although the dataset provides both RGB and NIR modalities, this study focuses exclusively on the RGB imagery. To ensure statistical reliability, we excluded categories with insufficient samples (fewer than 50 instances), such as planes, motorcycles, and buses. For the experimental setup, the selected data was randomly partitioned into a training set and a validation set with a ratio of 7:3.

B. Comparisons With Previous Methods

To comprehensively evaluate the effectiveness and robustness of the proposed SFBF-YOLO, we conduct experiments on three representative datasets: VisDrone-2019, USOD, and VEDAI. Specifically, all input images are uniformly resized to 640×640, and SFBF-YOLO is compared against the baseline YOLOv8 model. This setup allows us to preliminarily assess the generalization capability and robustness of our approach under diverse and challenging conditions.

1) VisDrone-2019: Table I presents a comprehensive evaluation of the proposed SFBF-YOLO on the VisDrone-2019 dataset, with the results succinctly visualized in Fig. 6. As illustrated in Fig. 6, our model substantially outperforms the baseline YOLOv8. The scatter points represent a variety of notable improved YOLO-based algorithms, real-time detectors, and two-stage detectors. Compared with the majority of these approaches, the yellow star markers (SFBF-YOLO) exhibit markedly lower model complexity (in terms of parameter count and GFLOPs) while achieving higher mAP_{50}. For a few competing methods—such as SOD-YOLO-m, SOD-YOLO-l, and REMDET-l—our model attains comparable detection accuracy but demonstrates a steeper slope in the figure, indicating superior overall efficiency. Fig. 7 shows the comparison of detection results between SFBF-YOLO and YOLOv8 in various complex scenarios of VisDrone-2019.

Comparison with Baseline YOLOv8: SFBF-YOLO achieves a comprehensive suppression of the YOLOv8 baseline across all scales. Instead of trading efficiency for accuracy, our model improves both simultaneously. Notably, *SFBF-YOLO-n* outperforms the larger YOLOv8s, achieving a higher mAP_{50} (0.41 versus 0.40) with only **10%** of the parameters (1.06M versus 11.1M) and **48%** of the GFLOPs. Similarly, **SFBF-YOLO-s** surpasses YOLOv8m with a **88%** parameter reduction and **46%** computation saving, while improving mAP_{50} by nearly 10%. This trend holds for the medium scale, where SFBF-YOLO-m outperforms YOLOv8l with significantly lower complexity. These results confirm that the proposed dual-stream reconstruction mechanism and SFEP are far more parameter-efficient than simply scaling up the network depth or width.

We selected three competitive versions of YOLO, namely YOLOv9-c, YOLOv10-s, and YOLOv11-s, for comparison. In fact, even other variants of YOLOv9, YOLOv10, and YOLOv11 do not outperform our model.

Comparison with improved YOLO algorithm: As further illustrated in Table I, our proposed model outperforms a series of improved YOLO-based detectors specifically designed for challenging aerial scenarios.

Despite competing against architectures featuring sophisticated modules, such as BiFormer (UAV-YOLO [41]), DCNv2 (EFA-Net [43]), contextual Transformers (YOLO-DCTI [45]), and specialized attention mechanisms (PC-YOLO11s [46], BDH-YOLO [44]), SFBF-YOLO maintains a clear performance advantage. It consistently achieves superior mAP_{50} and $mAP_{50\text{-}95}$, demonstrating its robustness and effectiveness for SOD in complex remote sensing scenarios.

Comparison With Other SOTA: In the domain of anchor-free object detection, RemDet [47] represents a recent SOTA model specifically designed for UAV imagery. RemDet addresses SOD from the perspective of information loss, positing that leveraging higher dimensional feature representations mitigates information degradation and consequently improves detection performance for small targets. In parallel, CEASC [48] proposes a global context-enhanced adaptive sparse

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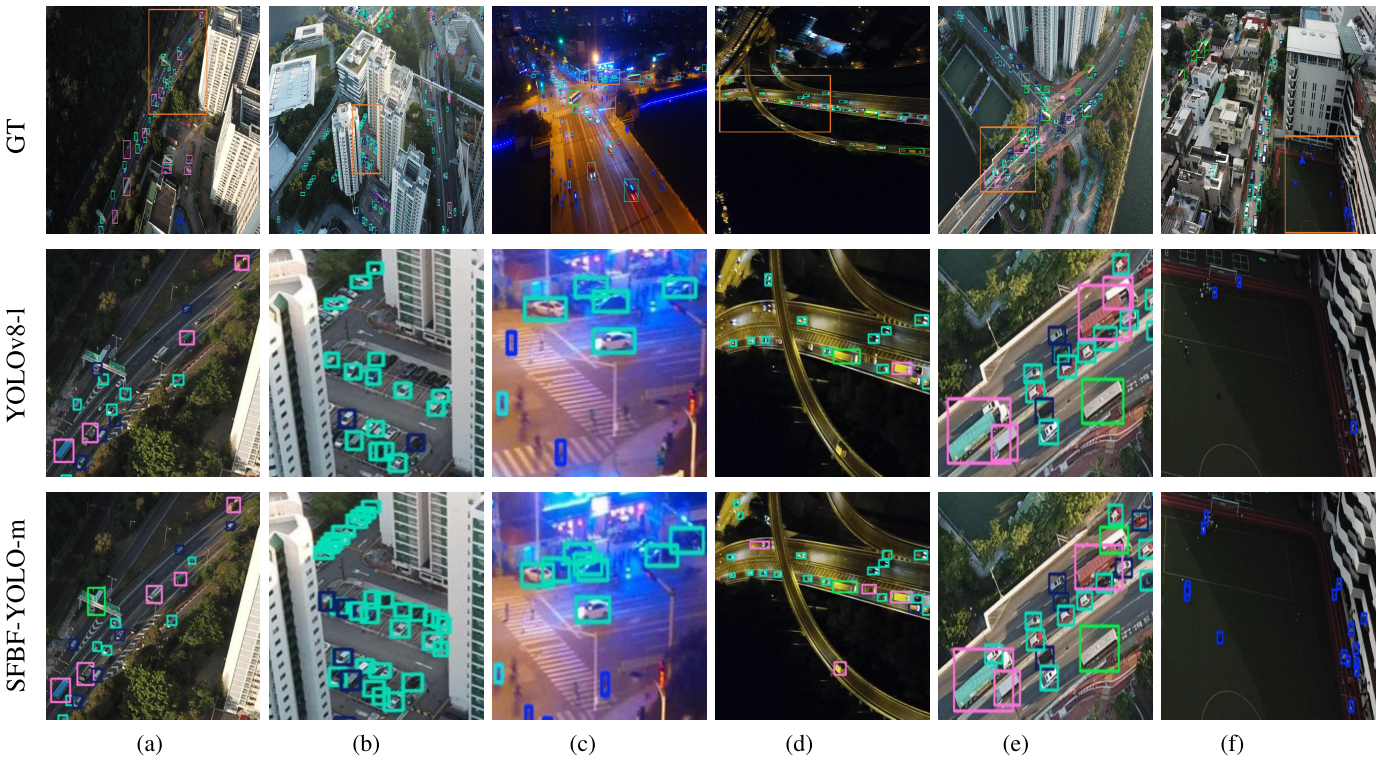


Fig. 7. Visual analysis of detection performance in challenging scenarios. (a)–(f) Common challenges, including uneven exposure, complex scenes with mutual occlusion, and dense target clusters. Specifically (a) overexposed high-rise buildings interfere with texture perception, (b) and (f) targets with minimal pixel proportions (yellow boxes) are densely clustered between buildings (b) and in low-light corners (f), respectively, (c) severe background light pollution, and (d) and (e) dynamic traffic scenes under night/day conditions. Notably, in the yellow-highlighted regions, YOLOv8 exhibits significant missed detections and false alarms, particularly failing in clustered targets and low pixel-ratio cases, whereas SFBF-YOLO demonstrates robust detection capabilities.

TABLE II
COMPARISON EXPERIMENTS FOR SFBF-YOLO IN USOD

Model	Param(M)	Gflops	mAP_{50}	mAP_{50-95}	precision	recall	FPS
SFBF-YOLO-s	**3.61**	42.6	0.905	0.346	0.912	0.852	159
FFCA-YOLO [30]	7.12	51.7	0.902	0.347	0.923	0.848	155
l-FFCA-YOLO	5.04	37.6	0.89	0.336	0.92	0.814	165
FFCA-YOLO w/ v8 Head	8.45	71.7	0.911	0.351	0.913	0.872	118
L-FFCA-YOLO w/ v8 Head	6.38	58.3	0.894	0.333	0.907	0.86	123
TPH-YOLOv5 [49]	45.36	252	0.891	0.32	0.902	0.844	68
Gold-YOLO [50]	26.75	52.9	0.872	0.316	0.877	0.827	89
SOD-YOLO-l	17.6	167	0.909	0.347	0.918	0.864	100
YOLOv8m	25.9	79.1	0.865	0.311	0.879	0.813	96
YOLOv5m	21.2	49	0.859	0.309	0.871	0.811	110

convolutional network tailored for UAV-based object detection. Despite the architectural innovations and performance advantages demonstrated by both RemDet and CEASC, our proposed SFBF-YOLO achieves superior overall performance in terms of both detection accuracy and computational efficiency, highlighting its robustness and efficacy for SOD in complex remote sensing environments.

2) USOD: As reported in Table II, the proposed SFBF-YOLO-s achieves a remarkable inference speed of 159 FPS while maintaining top-tier precision (mAP_{50} of 0.905).

To ensure a rigorous and comprehensive evaluation, we benchmarked our method not only against the standard FFCA-YOLO cited from its original publication [30], where USOD was introduced, but also further explored a modified variant. In this exploratory setting, we replaced the original YOLOv5-based detection head of FFCA-YOLO with that of YOLOv8 (w/ v8 Head), aligning the detection head configuration with SFBF-YOLO to investigate the performance limits under a unified head architecture.

The results show that SFBF-YOLO-s outperforms the standard FFCA-YOLO in both accuracy (0.905 versus 0.902) and with only 50.7% of the parameters (3.61M versus 7.12M).

Even when FFCA-YOLO is equipped with the advanced YOLOv8 head, which marginally improves its accuracy to 0.911, it suffers from increased computational cost. In contrast, SFBF-YOLO-s maintains comparable accuracy (within 0.06 difference) and higher FPS while utilizing only 42.7% of the parameters and 59.4% of the GFLOPs.

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TABLE III
COMPARISON EXPERIMENTS FOR SFBF-YOLO IN VEDAI

Model	Boat	Camping Car	Car	Pickup	Tractor	Truck	Vans	Others	mAP_{50}	FPS
SFBF-YOLO-s	0.47	0.653	0.829	0.713	0.612	0.591	0.702	0.408	0.622	151
FFCA-YOLO	0.365	0.595	0.844	0.685	0.647	0.626	0.606	0.386	0.594	99
L-FFCA-YOLO	0.401	0.592	0.818	0.701	0.631	0.681	0.673	0.334	0.604	101
YOLOv8m	0.388	0.599	0.799	0.667	0.621	0.622	0.62	0.457	0.597	123
UAV-YOLO	0.322	0.665	0.822	0.736	0.612	0.559	0.553	0.391	0.582	83
TPH-YOLO	0.262	0.585	0.752	0.666	0.542	0.489	0.483	0.321	0.512	79
SOD-YOLO-m	0.472	0.674	0.835	0.736	0.592	0.564	0.615	0.374	0.608	105

TABLE IV
OVERALL ABLATION EXPERIMENT

SR	BPAN	SFGA	FGSF	PB	Param(M)	Gflops	mAP_{50}	mAP_{50-95}
					3.0	8.1	0.341	0.195
√					0.99	10.4	0.373	0.219
√	√				0.99	10.4	0.378	0.221
√		√			1.09	13.9	0.416	0.245
√		√	√		1.03	13.1	0.402	0.238
√		√	√	√	1.06	13.7	0.408	0.242
√	√	√	√	√	1.06	13.7	0.410	0.243

Collectively, the results in Table II underscore the exceptional generalization capability and robustness of the proposed method. These findings suggest that the architectural improvements are not confined to the VisDrone benchmark but can effectively transfer to other challenging SOD tasks across diverse remote sensing environments.

Crucially, the comparative analysis against the structurally aligned FFCA-YOLO w/ v8 Head isolates the source of these efficiency gains. It empirically validates that the superior performance stems intrinsically from the proposed dual-stream reconstruction mechanism and the SFEP, rather than merely the adoption of an advanced detection head.

3) VEDAI: To further corroborate the robustness and generalization capability of the proposed method across diverse aerial scenes, quantitative results on the VEDAI dataset are summarized in Table III. Specifically, it attains a leading mAP_{50} of 0.622, outperforming SOTA competitors, such as SOD-YOLO and L-FFCA-YOLO, while maintaining a high inference speed of 151 FPS, significantly faster than both lightweight and heavy counterparts.

For categories with high-interclass similarity, such as vans (AP = 0.702) and pickups (AP = 0.713), which are frequently misclassified as standard cars in nadir-view imagery, our method achieves exceptional classification precision. This empirically confirms that the granularity stream in SFGA successfully locks onto the subtle intrinsic geometric signatures required for fine-grained distinction.

In the boat category, where detection is complicated by varying water reflections and dry-dock backgrounds, SFBF-YOLO-s maintains robust performance (AP = 0.47), rivaling the SOTA SOD-YOLO. This suggests that the Contextual Stream effectively captures the extrinsic environmental interactions, preventing the target signal from being overwhelmed by background clutter.

Overall, the balanced performance across both common and challenging categories underscores the method's strong operational versatility for comprehensive aerial surveillance.

TABLE V
COMPARISON EXPERIMENTS FOR BPAN IN VISDRONE-2019

Prop.strategy	Param(M)	Gflops	mAP_{50}	mAP_{50-95}
PAN	0.99	10.4	0.373	0.219
BPAN-a	0.99	10.4	0.366	0.212
BPAN-b	0.99	10.4	0.374	0.218
BPAN-c	0.99	10.4	0.372	0.219
BPAN-d	0.99	10.4	0.371	0.217
BPAN-e	0.99	10.4	**0.378**	**0.221**
AFPN [51]	1.06	11.8	0.360	0.210
RepGFPN(DAMO-YOLO) [52]	1.09	10.8	0.376	0.220
BiFPN [53]	0.99	10.5	0.365	0.213
PAN(w Dysample [14])	1.05	10.7	0.378	0.222

C. Ablation Study

To evaluate the effectiveness of each component in SFBF-YOLO, we progressively integrated SR, BPAN, SFGA, FGSF, and PB into the baseline to verify their individual contributions. All experiments were conducted on the VisDrone-2019 dataset. Table IV summarizes the impact of adding or removing each module on the evaluation metrics, where √ indicates that the corresponding module is included, and a blank indicates that it is not used.

1) SR: SR represents the structural reconstruction of the framework, where we removed the low-contribution P5 feature representation from the backbone and incorporated a more small-object-sensitive P2 feature representation into the neck.

As shown in Table IV, after the structural reconstruction, the number of parameters was reduced by 67%. However, due to the introduction of the high-resolution P2 feature map in the neck, the GFLOPs increased by 25%. Meanwhile, a significant improvement in detection performance was observed, with mAP_{50} increasing from 0.341 to 0.373 and $mAP_{50\text{-}95}$ increasing from 0.195 to 0.219.

2) BPAN: Integrating BPAN further improved mAP_{50} to 0.378 without adding parameters, proving the efficiency of our feature fusion strategy.

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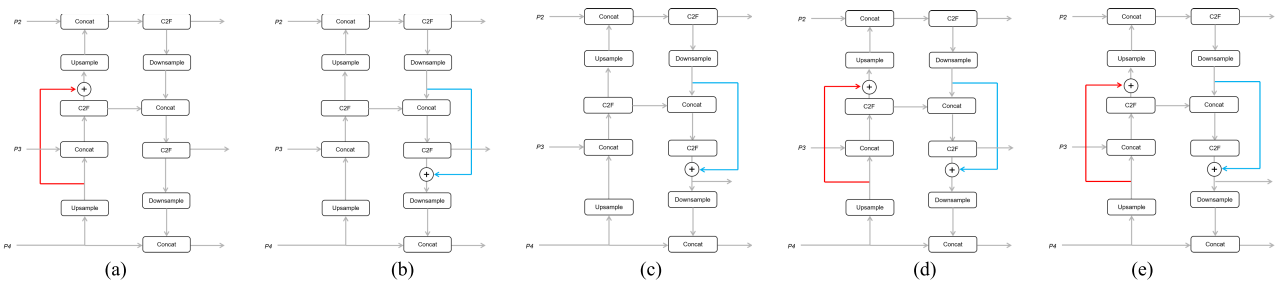


Fig. 8. Implementation strategies of BPAN. (a) Residual connection from P4 to P3 before P3-to-P2 fusion in the top-down path to supplement P2 with P4 information. (b) Residual connection from P2 to P3 before P3-to-P4 fusion in the bottom-up path to supplement P4 with P2 information. (c) Extending (b) by sending the fused P4–P3 residual output as the final P3 representation to the detection head. (d) Combination of (a) and (b). (e) Combination of (a) and (c).

TABLE VI

ABLATION STUDY OF SFGA INTEGRATION AT DIFFERENT FEATURE LEVELS ON VISDRONE-2019. "1" DENOTES THE INTEGRATION OF THE SFGA MODULE AT THE CORRESPONDING P-LEVEL, WHILE "0" INDICATES USING THE BASELINE SPD-S CONFIGURATION

#	P2	P3	P4	mAP_{50}	$mAP_{50:95}$
1	1	1	1	0.413	0.246
2	1	0	0	0.407	0.242
3	0	1	0	0.403	0.237
4	0	0	1	0.404	0.239
5	1	0	1	0.41	0.243
6	0	1	1	0.407	0.241
7	1	1	0	**0.416**	0.245

To validate the proposed *SFEP*, we explored different interaction strategies, as illustrated in Fig. 8 and Table V. Strategies (a-d) represent unidirectional or partially compensated interactions. As observed, merely supplementing P2 with semantic P4 (Strategy a) or P4 with spatial P2 (Strategy b) fails to achieve optimal results due to the *information decoupling* for enhancing one aspect often dilutes the other.

In contrast, Strategy (e) (our BPAN) establishes a fully symmetric equilibrium. It effectively leverages the enriched semantic information to mitigate semantic dilution when P2 supplements deeper layers, while simultaneously preserving localization precision. This balanced reciprocity yields the highest performance, confirming that multiscale interaction requires a holistic equilibrium rather than simple summation. Comparative experiments with AFPN [51], RepGFPN [52], and BiFPN [53] (see Table V) further demonstrate that BPAN achieves the superior feature integration efficiency.

3) SFGA: Incorporating the SFGA module yields the most significant performance leap (mAP_{50} from 0.373 to 0.416) with negligible overhead (+0.1M Params), as shown in Table IV.

Table VI indicates that a composite integration strategy covering both P2 and P3 levels yields the optimal gain. This finding refines our *dual-stream* theory: while P2 provides the foundational intrinsic physical signatures, extending the granularity locking to P3 ensures a progressive reinforcement of weak signals. This cascading preservation creates a robust signal safety corridor, preventing fine-grained details from being diluted during the critical transition from spatial-dominant to semantic-dominant layers. Table VII further confirms that SFGA outperforms other enhancement methods like FEM and RFCBAM.

TABLE VII

COMPARISON EXPERIMENTS FOR SFGA IN VISDRONE-2019. HERE, W/O DENOTES THE BASELINE MODEL THAT ONLY INTEGRATES THE SR WITHOUT THE SFGA MODULE. SPD-S INDICATES SPD [36] WITH CHANNEL SHUFFLE, AS DEFINED IN (1), ARE INCORPORATED INTO FEATURE LEVELS P2–P4 ON TOP OF THE W/O CONFIGURATION. FURTHERMORE, FEM [30] AND RFCBAM [13] ARE INCLUDED AS REPRESENTATIVE SOTA METHODS SPECIALIZED IN ENHANCING THE SEMANTIC CONTEXT AND FINE-GRAINED REPRESENTATION, RESPECTIVELY

Methods	Param(M)	mAP_{50}	mAP_{50-95}
w/o	0.99	0.373	0.219
SPD	1.05	0.382	0.225
SPD-S	1.05	0.398	0.232
FEM	0.96	0.351	0.201
RFCBAM	1.83	0.381	0.225
SFGA	1.09	**0.416**	**0.245**

TABLE VIII

EVALUATION OF FGSF'S LIGHTWEIGHT EFFECTIVENESS ACROSS DIFFERENT MODEL SCALES. W/O MEANS WITHOUT

Prop.strategy	Param(M)	Gflops	mAP_{50}	mAP_{50-95}
SFBF-YOLO-n(w/o FGSF)	1.12	14.5	0.421	0.247
SFBF-YOLO-n	1.06	13.7	0.41	0.243
SFBF-YOLO-s(w/o FGSF)	3.77	45.8	0.486	0.297
SFBF-YOLO-s	3.61	42.6	0.479	0.292
SFBF-YOLO-m(w/o FGSF)	11.2	118	0.515	0.317
SFBF-YOLO-m	9.52	100.6	0.503	0.307

Visual Analysis (see Fig. 9): Qualitative comparisons reveal that while standard SPD helps in highlighting structures, it struggles under high-contrast or cluttered conditions (e.g., Rows 1, 2, and 5). In contrast, SFGA consistently generates sharper activations with well-defined object boundaries and suppressed background noise. This confirms SFGA's ability to simultaneously lock intrinsic granularity and capture extrinsic context, ensuring robust feature reconstruction even in adverse lighting.

4) FGSF: As shown in Table IV, when incorporated independently on top of the SFGA-enhanced baseline, FGSF leads to a slight reduction in detection accuracy, with parameters decreasing from 1.09M to 1.03M, and Gflops from 13.9 to 13.1. This can be attributed to the potential information loss introduced by the compression mechanism of FGSF. To better understand the contribution of FGSF to model compactness, we further evaluate its impact on models of different scales, as shown in Table VIII. Across the nano, small, and medium configurations, the inclusion of FGSF consistently reduces the

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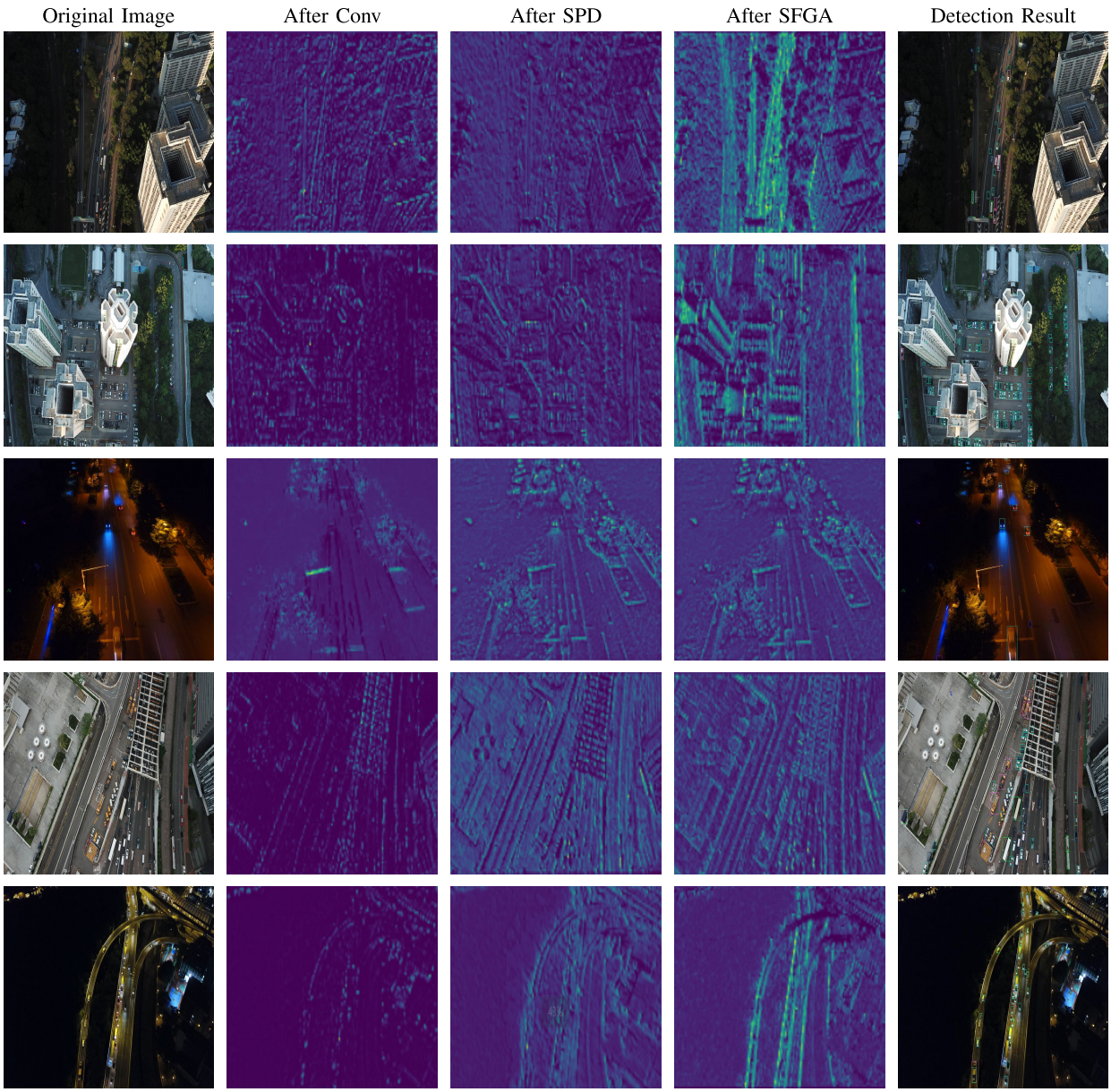


Fig. 9. Influence of SFGA on feature extraction. In the feature map visualization, higher regional contrast signifies a greater impact weight. After feature enhancement by the proposed SFGA module, the target's boundary region exhibits significantly heightened contrast and well-defined contours. It demonstrates that both the fine-grained details and high-level semantics of the target have been precisely and effectively enhanced.

parameter size and computational complexity. For instance, in the nano setting, FGSF reduces model size from 1.09M to 1.06M and computation from 13.9G to 13.1G, while maintaining strong performance. Similar trends are observed in the small and medium variants, where FGSF reduces computation by 3.2G and 17.4G, respectively, with only marginal drops in accuracy. These results clearly demonstrate the scalability and generalization of FGSF across different model sizes, proving it to be a lightweight yet effective enhancement module suitable for resource-constrained deployments.

5) PB: To compensate for the performance degradation caused by the FGSF module, we introduce the ParNet block (PB) as an enhancement component. When paired with the Parnet Block, pushing the $mAP_{50\text{-}95}$ back up to 0.242 and mAP_{50} to 0.408.

The full model configuration, which integrates SR, SFGA, FGSF, BPAN, and PB, eventually achieves the best tradeoff, delivering a competitive mAP_{50} of 0.41 and $mAP_{50\text{-}95}$ of 0.243 while maintaining a relatively low parameter count (1.06M) and computational cost (13.7GFLOPs).

D. Discussion on Limitations and Failure Cases

Despite the robust performance demonstrated in Fig. 7, where SFBF-YOLO excels in identifying tiny targets against relatively homogeneous backgrounds, a systematic analysis of failure cases reveals specific bottlenecks in extreme scenarios. As illustrated in Fig. 10, while the model maintains high precision for most objects, its detection reliability slightly degrades when encountering environmental shifts and high-density clustering.

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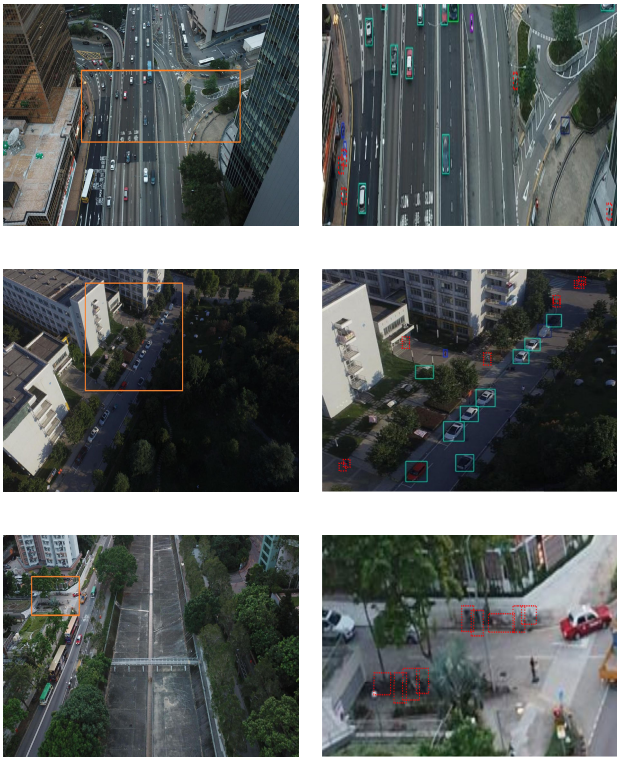


Fig. 10. Visual analysis of detection bottlenecks and failure cases. Red dashed boxes indicate missed detections (false negatives). While SFBF-YOLO handles isolated tiny targets effectively, it faces challenges in scenarios involving dense clustering, complex background clutter, and extreme environmental variations.

Specifically, in scenarios with complex background textures (e.g., urban shadows or vegetation clutter), the signal-to-noise ratio (SNR) of ultratiny targets is significantly reduced. This makes it difficult for the SFGA module to extract sufficient discriminative features from a handful of pixels. Furthermore, in highly crowded regions—such as dense parking lots or urban intersections—mutual occlusion between targets leads to spatial ambiguity. In these cases, even with the bidirectional support of BPAN, the refined P2 feature map occasionally fails to resolve individual instances that are tightly packed or partially obscured by natural elements. These observations suggest that although the synergistic integration of SFGA and BPAN significantly enhances the model's capability in preserving fine-grained details, contextual awareness, and balancing multiscale information, achieving the absolute robustness under extreme environmental interference and ultrahigh target density remains an inherent challenge. The persistent bottlenecks in these extreme cases indicate that even with advanced spatial-semantic coordination, the physical limitations of signal-to-noise ratio and spatial ambiguity continue to define the operational boundaries of current detectors. Future research may explore the integration of temporal consistency or adaptive NMS strategies to further mitigate these limitations.

V. Conclusion

In this article, we investigated the fundamental physical bottleneck of fine-grained information degradation and representation conflict in UAV-based SOD. Unlike conventional approaches that rely on heuristic feature stacking, we argued that addressing these constraints requires a fundamental transition toward source-level information reconstruction and symmetric feature modeling.

To this end, we presented SFBF-YOLO, a unified algorithm grounded in two methodological pillars.

1) *Dual-Stream Reconstruction:* We demonstrated that the information degradation at the network source is not strictly inevitable. By introducing the SFGA module, we validated that bifurcating feature extraction into a granularity stream and a contextual stream enables the effective recovery of weak signals that are typically lost in standard downsampling.
2) *Symmetric Feature Equilibrium:* We established the SFEP principle instantiated by BPAN. Our results confirm that multiscale interaction should be modeled through a bidirectional reciprocity logic rather than simple fusion. This guarantees that the feature representations avoid functional imbalance caused by a polarized emphasis on either localization-oriented precision or recognition-oriented discriminability.

Furthermore, the integration of the FGSF module proved that high-dimensional feature evolution can be achieved with high efficiency through redundancy-aware mining. Extensive evaluations on VisDrone-2019, USOD, and VEDAI benchmarks empirically validated our hypothesis: granularity and context preservation at the source and symmetric equilibrium in the neck are the two critical prerequisites for robust SOD.

In future work, we plan to generalize these design principles to cross-domain adaptive detection, exploring how the proposed dual stream reconstruction mechanism and SFEP can robustly handle unseen environmental domains (e.g., nighttime or adverse weather) without extensive retraining, thereby further enhancing the operational versatility of UAV and remote sensing perception systems.

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